

between people (users) given their characteristics and preferences. For this, the system/machine needs to convert people into mathematical data. This data collection is usually done by gathering details through common auto-authentication accounts such as Facebook or Google. Then that data is enhanced, usually by using a detailed questionnaire. The questions are designed to measure various personality vectors that are proprietary to each company. Personalities are then tested for compatibility subjecting this multi-dimensional data to relevant queries (running the algorithm).

Now every company will have their own way of trying to be more accurate, but the basic method is pretty much the same. The biggest flaw of this system is that the information entered is usually wrong. The problem with us humans is that we perceive ourselves to be something we're usually not. When self reporting, we are biased. We provide data about who we think we are, or often who we would like to be. We over- or under-estimate different aspects of our personalities, and too

often we just give the answers we think a potential match would want to see.

Add in the complexity of humans and how we cannot all be fitted into neat little categories (no matter how hard the companies try), and you end up with a really bad description of the "real you" in a dating website's dataset. The only leveller might be that your potential match's data is probably just as bad a descriptor of them, so you could still potentially hit it off!

OkCupid tries to solve for this by asking questions like "Do you like to discuss politics with your partner?" or "Are you annoying?". The only truthful answer to this is "It depends", but how many people will say that?

The first time computer algorithms were used to match people was back in 1965, when computers still used punch cards. Jeff Tarr, then a mathematics student at Harvard, gave out questionnaires for people to describe their "ideal date". The information was then processed in an IBM-1401 according to his algorithm and six suggestions were spit out for a given person (which was then

mailed to them. Each of the six suggestions also got the given person's contacts along with that of five others. It turned out to be a success and a year later Tarr boasted about having received over 90,000 applications for Operation Match. Even then, there was something about the aura of a



Thank Algorithms we're together!

match making machine that attracted people who bestowed a special faith in the decisions of this machine.

We have come a long way since then, in terms of the mathematics of the algorithms as well as the basis of the questionnaires. Academic style observation certainly helps. eHarmony's algorithm was the first to be based on scientifically conducted research (by psychologist Neil Clark Warren, who collected over 3 years of data from 5,000 couples). It's a self-fulfilling cycle from then on, as more user data leads to better matches and hence more users, and so on. For the other category of dating apps that don't suggest matches, the mathematics is purely based on a one-is-to-one profile assessment and if it is a two way approval, users are matched. After the category and some metadata is specified, the onus is on the user to discriminate between choices. This system is easily gamed, however.

Chris McKinlay, a math student at UCLA, managed to group categories of OkCupid's user base and figure out how the algorithms match amongst them. Then he adjusted his profile according to the two groups he wanted to match with, and voila! There were 10-12 girls texting him every day. He owned their system, and went on a dating spree, a year before he married his 88th date. All that math and it still

SEXFIT

As gamers, we are already familiar with the idea of extracting statistics such as Actions Per Minute (or APMs) from our gameplay. APM for example, gives an indicator of how many clicks and keystrokes a gamer makes within a minute. This stat is often regarded as the primary unit for measuring a gamer's skill especially in RTS games, with pro-gamers easily reaching upwards of 350 APMs at crucial moments in the game. It is only natural, then, that geeks like us would want to track performance in err... other areas. In fact, this could be the ultimate pinnacle of achievement for wearable tech – tracking your TPMs (Thrusts Per Minute)!

The device packs in an accelerometer and can generate vital in-out data. With a bluetooth connected app, in theory, you could even have leaderboards. Think about it, geeks are good at everything that we do because we analyse data in the pursuit of efficiency. Now with something like SexFit you'll have hard data to back up your claims or get better. Too bad British adult retailer Bondara never really followed through with their concept. Sadly even three years since it was first conceptualised, it still remains a work-in-progress.



A fitness band of a different kind.